

Olive oil and walnut breakfasts reduce the postprandial inflammatory response in mononuclear cells compared with a butter breakfast in healthy men.

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BACKGROUND: Inflammation is crucial in all stages of atherosclerosis, and few studies have investigated the effect of dietary fat on markers of inflammation related to this disease during the postprandial period. OBJECTIVE: To evaluate the chronic effects of dietary fat on the postprandial expression of proinflammatory genes in peripheral blood mononuclear cells (PBMCs) in healthy subjects. DESIGN: 20 healthy men followed three different diets for 4 weeks each, according to a randomized crossover design: Western diet: 15% protein, 47% carbohydrates (CHO), 38% fat (22% saturated fatty acid (SFA)); Mediterranean diet: 15% protein, 47% CHO, 38% fat (24% monounsaturated fatty acid (MUFA)); CHO-rich and n-3 diet: 15% protein, 55% CHO, <30% fat (8% polyunsaturated fatty acid (PUFA)). After 12-h fast, volunteers were given a breakfast with a fat composition similar to that consumed in each of the diets-butter breakfast: 35% SFA; olive oil breakfast: 36% MUFA; walnut breakfast: 16% PUFA, 4% alpha-linolenic acid (LNA). RESULTS: The butter breakfast induced a higher increase in tumor necrosis factor (TNF)-alpha messenger RNA (mRNA) expression than the olive oil or walnut breakfasts ($P=0.014$) in PBMCs. Moreover, we found a higher postprandial response in the mRNA of interleukin (IL)-6 with the intake of butter and olive oil breakfasts than with the walnut breakfast ($P=0.025$) in these cells. However, the effects of the three fatty breakfasts on the plasma concentrations of these proinflammatory parameters showed no significant differences ($P=N.S.$). CONCLUSION: Consumption of a butter-enriched meal elicits greater postprandial expression of proinflammatory cytokine mRNA in PBMCs, compared to the olive oil and walnut breakfasts.